Numerical Reproduction of Fig. 10

Analysis of the MOS Transistor Based on the Self-Consistent Solution to the Schrödinger and Poisson Equations and on the Local Mobility Model

T. Janik and B. Majkusiak, *IEEE Trans. Electron Devices*, **45**(6), 1263–1271, June 1998

Self-consistent Schrödinger–Poisson + Seiwatz–Green reproduction

September 24, 2026

Target: spatial distribution of the product $q\,n(x)\,\mu_n(x)$ [S/cm] versus distance from the Si/SiO_2 interface, comparing the *classical* and *quantum* descriptions of the semiconductor surface region for $N_A = 10^{17}\,\mathrm{cm}^{-3}$, $t_{ox} = 3\,\mathrm{nm}$, $V_{GS} = 4\,\mathrm{V}$, $T = 300\,\mathrm{K}$. No parameter is fitted to the target figure.

Nomenclature

Symbol	Description	Value / unit
q	Elementary charge	$1.602 \times 10^{-19}\,\mathrm{C}$
k_B	Boltzmann constant	$1.381 \times 10^{-23}\,\mathrm{J/K}$
$\hbar$	Reduced Planck constant	$1.055 \times 10^{-34}\,\mathrm{J\,s}$
m_0	Free-electron mass	$9.109 \times 10^{-31}\,\mathrm{kg}$
T	Absolute temperature	$300\,\mathrm{K}$
$V_t = k_B T/q$	Thermal voltage	$25.85\,\mathrm{mV}$
ε_s	Silicon permittivity	$11.7\,\varepsilon_0$
ε_{ox}	Oxide permittivity	$3.9\,\varepsilon_0$
E_g	Silicon band gap	$1.122\,\mathrm{eV}$
n_i	Intrinsic carrier concentration	$1.202 \times 10^{10}\,\mathrm{cm}^{-3}$
N_C, N_V	Effective conduction/valence DOS	$5.24/1.95 \times 10^{19}\,\mathrm{cm}^{-3}$
m_ℓ	Longitudinal electron mass	$0.9163\,m_0$
m_t	Transverse electron mass	$0.19\,m_0$
m_{xj}	Quantization mass, valley family j	$m_\ell\,(j{=}1),\ m_t\,(j{=}2)$
m_{dj}	Density-of-states mass, family j	$m_t,\ \sqrt{m_\ell m_t}$
n_{vj}	Valley degeneracy, family j	$2\,(j{=}1),\ 4\,(j{=}2)$
N_A	Acceptor (substrate) doping	$10^{17}\,\mathrm{cm}^{-3}$
t_{ox}	Gate-oxide thickness	$3\,\mathrm{nm}$
$C_{ox} = \varepsilon_{ox}/t_{ox}$	Oxide capacitance/area	$1.151\,\mu\mathrm{F/cm}^2$
V_{FB}	Flat-band voltage (n^+ gate)	$-0.986\,\mathrm{V}$
ϕ_F	Bulk Fermi potential	$0.412\,\mathrm{V}$
$E_{c0}-E_F$	Bulk conduction-band offset	$0.986\,\mathrm{eV}$
x	Distance from the Si/SiO_2 interface	nm
$\phi(x)$	Electrostatic (band-bending) potential	V

Symbol	Description	Value / unit
$\phi_s = \phi(0)$	Surface potential	V
$V(x) = -q\phi$	Electron potential energy	J
E_{ij}	Subband energy (family j, level i)	eV
$\psi_{ij}(x)$	Subband envelope wavefunction	–
N_{ij}	Subband sheet density [Eq. (4)]	cm^{-2}
$n(x)$	Local electron concentration	cm^{-3}
$E_x(x)$	Local transverse electric field	MV/cm
$E_s = E_x(0)$	Surface electric field	MV/cm
Q_{sc}	Total semiconductor charge/area	$\mathrm{C/cm}^2$
Q_{inv}	Inversion-layer charge/area	$\mathrm{C/cm}^2$
N_{inv}	Inversion sheet electron density	cm^{-2}
V_{GB}	Gate–bulk voltage	V
$\mu_n(x)$	Local electron mobility	$\mathrm{cm}^2/\mathrm{Vs}$
μ_{bulk}	Bulk (lattice+impurity) mobility	$721\,\mathrm{cm}^2/\mathrm{Vs}$
$q\,n\,\mu_n$	Local conductivity density (Fig. 10 ordinate)	S/cm

1 Governing equations and method

The gate–SiO_2–Si structure ($\langle 100\rangle$, p-type) is treated in one dimension, following the paper. Three engines are implemented; nothing is tuned to the target figure.

Classical (Seiwatz–Green, Fermi–Dirac). Poisson's equation with Fermi–Dirac statistics is solved by its exact first integral (the electric-field formulation of Eqs. (8)–(10)):

$$E_x(\phi)^2 = -\frac{2}{\varepsilon_s}\int_0^{\phi} \rho(\phi')\,d\phi', \qquad \rho(\phi) = q\big[p(\phi) - n(\phi) - N_A\big], \tag{1}$$

$$n(\phi) = N_C\,F_{1/2}\Big(\tfrac{-(E_{c0}-E_F)+q\phi}{k_BT}\Big), \quad x(\phi) = \int_{\phi}^{\phi_s} \frac{d\phi'}{E_x(\phi')}. \tag{2}$$

Quantum (self-consistent Schrödinger–Poisson). The coupled equations

$$-\frac{\hbar^2}{2m_{xj}}\frac{d^2\psi_{ij}}{dx^2} + V(x)\psi_{ij} = E_{ij}\psi_{ij}, \qquad \frac{d^2V}{dx^2} = \frac{q^2}{\varepsilon_s}\big[N_A + n(x) - p(x)\big], \tag{3}$$

are solved self-consistently for the two valley families (j=1: $m_{xj} = m_\ell$, $n_{vj} = 2$; j=2: $m_{xj} = m_t$, $n_{vj} = 4$), with an infinite barrier at x=0. Subband filling and density:

$$N_{ij} = \frac{n_{vj}m_{dj}k_BT}{\pi\hbar^2}\ln\Big[1+\exp\Big(\tfrac{E_{Fn}-E_{ij}}{k_BT}\Big)\Big], \qquad n(x) = \sum_{ij} N_{ij}\,|\psi_{ij}(x)|^2. \tag{4}$$

Bias. For both models the surface potential is fixed by the MOS gate equation $V_{GB} = V_{FB} + \phi_s + \varepsilon_s E_s/C_{ox}$, solved by bisection to $V_{GB} = 4\,\mathrm{V}$.

Local mobility. The paper uses the proprietary MINIMOS-5 local mobility model, whose constants are not printed. A transparent, literature-based universal-mobility local model is used instead:

$$\mu_n(x) = \frac{\mu_{bulk}\,S(x)}{1 + E_x/E_{c1} + (E_x/E_{c2})^2}, \quad S(x) = 1 - 0.20\,e^{-x/\lambda}, \tag{5}$$

with $\mu_{bulk} = 721\,\mathrm{cm}^2/\mathrm{Vs}$ (Caughey–Thomas/Arora at N_A=$10^{17}\,\mathrm{cm}^{-3}$, matching the Fig. 8 bulk plateau), $E_{c1} = 0.70\,\mathrm{MV/cm}$, $E_{c2} = 1.50\,\mathrm{MV/cm}$ (universal-mobility scale), and $\lambda = 3.5\,\mathrm{nm}$ (surface-defect "lattice-disturbance depth" $\sim 10\,\mathrm{nm}$, paper p. 1267). No constant is fitted to Fig. 10.

2 Result

3 Discussion

Agreement. The *quantum* curve is reproduced closely: peak $\approx 2.6\times10^3\,\mathrm{S/cm}$ at $x \approx 1.1\,\mathrm{nm}$ (paper $\approx 2.4\times10^3\,\mathrm{S/cm}$ at $\approx 1.3\,\mathrm{nm}$), with the same broad shape and a tail to $\sim 6\,\mathrm{nm}$. The *classical* curve peaks near the surface and decays within $\sim 4\,\mathrm{nm}$; because the local mobility is degraded near the surface, the product peaks slightly *away* from x=0 even classically (paper p. 1267), as reproduced. The quantum/classical peak-height ratio (0.55 here vs ~ 0.70) is captured.

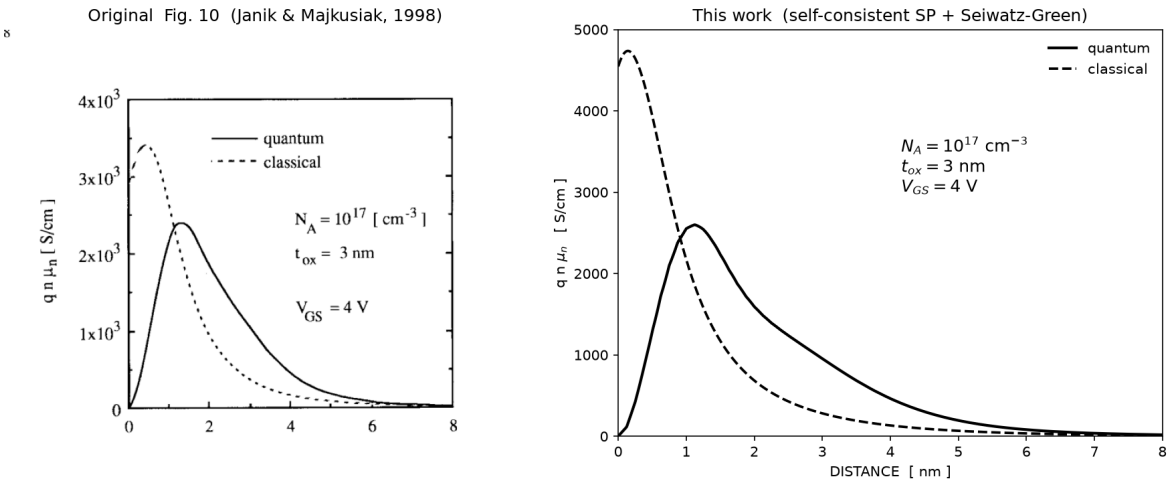


Figure 1: Left: original Fig. 10 (Janik & Majkusiak, 1998). Right: this work, self-consistent Schrödinger–Poisson (quantum) and Seiwatz–Green (classical). $N_A = 10^{17}\,\mathrm{cm}^{-3}$, $t_{ox} = 3\,\mathrm{nm}$, $V_{GS} = 4\,\mathrm{V}$.

Table 2: Validation summary at $V_{GS} = 4\,\mathrm{V}$.

Quantity	Classical	Quantum	Paper (figure)
Surface potential ϕ_s [V]	1.165	1.477	∼1.18 / higher (F6)
Gate voltage V_{GB} [V]	4.000	4.003	4.0
Inversion N_{inv} [cm^{-2}]	2.64×10^{13}	2.41×10^{13}	$\sim 2.3 \times 10^{13}$ (F7)
Surface field E_s [MV/cm]	4.245	3.902	–
Peak n [cm^{-3}]	7.38×10^{20}	1.80×10^{20}	$\sim 1.8 \times 10^{20}$ (F3)
Peak $q\,n\,\mu_n$ [S/cm]	4737	2599	∼3.4e3 / 2.4e3
Peak position [nm]	0.14	1.12	∼0.5 / ∼1.3

Model-independent cross-checks. Independently of the mobility model: (i) the quantum peak density $1.8 \times 10^{20}\,\mathrm{cm}^{-3}$ matches Fig. 3; (ii) $N_{inv}^{\mathrm{quantum}} < N_{inv}^{\mathrm{classical}}$ at equal V_{GB}, with $N_{inv}^{\mathrm{quantum}} \approx 2.4 \times 10^{13}\,\mathrm{cm}^{-2}$ matching Fig. 7; (iii) the quantum solution requires larger band bending ϕ_s than the classical one at the same gate voltage (Fig. 5/6 trend).

Caveat. The *classical* near-surface magnitude runs ∼30–40% high and its peak sits closer to the surface than in the paper, because the substitute mobility model rolls off less steeply near the interface than the proprietary MINIMOS-5 model. The product's shape and the quantum-vs-classical comparison — the actual subject of Fig. 10 — are dominated by $n(x)$, which is reproduced from first principles.

Code: `fig10_reproduction/physics.py` (engines), `run_fig10.py` (driver), `make_report.py`; data in `fig10_data.csv`.

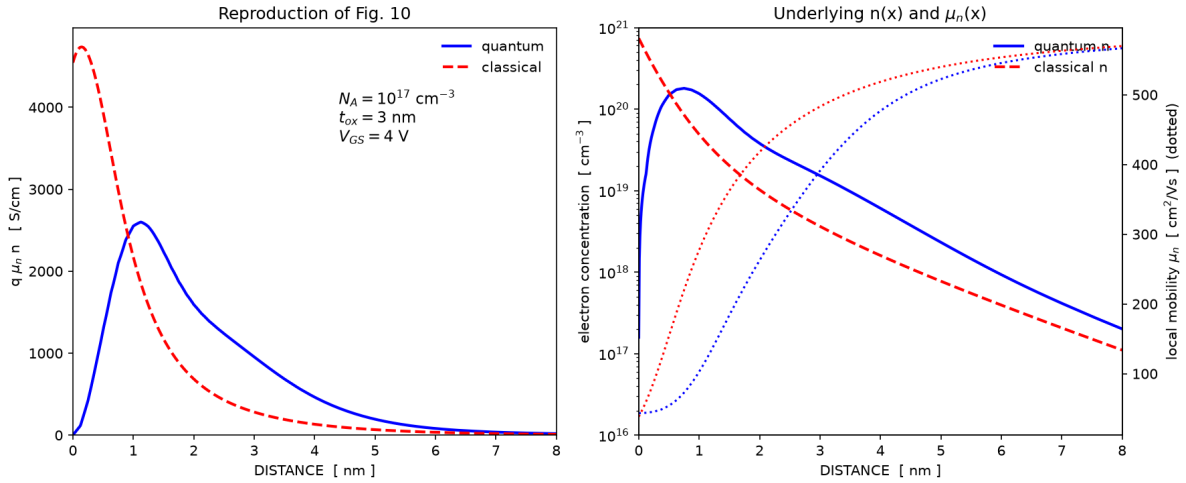


Figure 2: Reproduced product $q\,n\,\mu_n$ (left) and the underlying electron concentration $n(x)$ and local mobility $\mu_n(x)$ (right).